MEDIKIOSK CLINICAL PLATFORM DOCUMENTATION

DATASETS

Datasets Documentation

Last Analyzed: 2026-09-16 Commit:`f118085510bdf2fa44785ee3ca5aaf5b4875c209`

Dataset Inventory Summary

The repository includes, processes, or references three distinct dataset collections covering clinical OPD dialogues, traditional Ayurvedic text collections, and ICD-10 medical code mappings.

1. OPD Clinical Dialogues Dataset (Synthetic)

- **Name**: Indian OPD Clinical Intake & Triage Dialogues
- **Type**: Synthetic Text (JSONL format)
- **Source / URL**: Local repo file `training/opd_dialogues.jsonl`
- **Version / Hash**: TODO: verify commit snapshot
- **License**: TODO: verify repository license (MIT)
- **Size**: 1.46 KB (Sample demonstration set)
- **Format**: JSON Lines (`.jsonl`)
- **Local Path**: `training/opd_dialogues.jsonl`
- **Download / Access Instructions**: Included directly in repository.
- **Preprocessing Steps**: Formatted into Llama 3 ChatML / instruction prompt structure via `format_prompt()` in `training/qlora_finetune.py:L58-L69`.
- **Train / Validation / Test Splits**:
  - Train: 100% (Demonstration split loaded via `load_dataset("json", data_files="opd_dialogues.jsonl", split="train")`)
  - Validation / Test: None
- **Columns / Features / Labels**:
  - `system`: System instruction defining clinical role & triage guidelines.
  - `user`: Patient chief complaint and symptom inputs.
  - `assistant`: Structured clinical assistant triage response.
- **Target Variable**: `assistant` text (causal language modeling target).
- **Where Used in Code**: `training/qlora_finetune.py:L21`, `training/qlora_finetune.py:L77`.
- **Model / Task it Supports**: Meta-Llama-3-8B QLoRA fine-tuning for Indian hospital OPD clinical history taking.
- **Citation**: Synthetic dataset generated for hackathon demonstration.
- **Privacy / Ethical Notes**: 100% synthetic clinical data; contains zero real patient Personally Identifiable Information (PII) or Protected Health Information (PHI).

2. Ayurveda / Traditional Medicine to ICD-10 Bridge Dataset

- **Name**: `AmareshHebbar/ayurveda-icd-sft` (AxisMapper Medical AI Suite)

- **Type**: Text (JSONL format)
- **Source / URL**: Hugging Face Dataset - AmareshHebbar/ayurveda-icd-sft
- **Version / Hash**: CMS FY2026 ICD-10-CM release baseline
- **License**: Apache 2.0 (`rag_corpus/raw/ayurveda_icd/README.md:L2`)
- **Size**: 1.64 MB total across splits (3,002 rows)
- **Format**: JSON Lines (`.jsonl`)
- **Local Path**: `rag_corpus/raw/ayurveda_icd/`
  - `train.jsonl` (1.31 MB)
  - `validation.jsonl` (162 KB)
  - `test.jsonl` (165 KB)
- **Download / Access Instructions**:

```python
from datasets import load_dataset
ds = load_dataset("AmareshHebbar/ayurveda-icd-sft")
```

- **Preprocessing Steps**: Extracted from public CMS ICD-10-CM Z-codes (health status) and Y-codes (external causes) mapped to traditional Indian medicine terms (Ayurveda, Unani, Siddha, Homeopathy). Formatted as ChatML `messages` list.
- **Train / Validation / Test Splits**:
  - Train: 2,401 rows
  - Validation: 300 rows
  - Test: 301 rows
  - Total: 3,002 rows
- **Columns / Features / Labels**:
  - `messages`: List of chat objects `[{"role": "system", "content": "..."}, {"role": "user", "content": "..."}, {"role": "assistant", "content": "..."}]`
- **Target Variable**: Assistant message containing ICD-10-CM code, description, and clinical context notes.
- **Where Used in Code**: Parsed by `backend/src/rag/parse.ts` and indexed into vector storage for RAG retrieval during Ayush clinical reasoning.
- **Model / Task it Supports**: Integrative medicine clinical documentation, ABDM ABHA health records integration, and ICD-10 automated coding.
- **Citation**:

```bibtex
@misc{axiomapper2026,
  author    = {Hebbar, Amaresh},
  title     = {AxisMapper: Medical AI Fine-tuning Dataset Suite},
  year      = {2026},
  publisher = {HuggingFace},
  url       = {https://huggingface.co/collections/AmareshHebbar/axiomapper-medical-ai-suite}
}
```

- **Privacy / Ethical Notes**: Extracted strictly from public CMS guidelines and medical coding reference standards; no personal health records used.

3. Indian Vedas and Ayurveda Corpus

- **Name**: Indian Vedas and Ayurveda Continuous Pre-Training Corpus

- **Type**: Multilingual Text (Sanskrit `sa`, English `en`, Hindi `hi`)
- **Source / URL**: Multi-source digital humanities collections (Griffith, Keith, Gulabkunverba Ayurvedic Society, Bhishagratna, Mookerji, IRJAY journal articles)
- **Version / Hash**: Public domain digital translations
- **License**: Public Domain / Open Educational (`license_name: public-domain` in `rag_corpus/raw/vedas/README.md:L3`)
- **Size**: 39.51 MB (`continueousPreTrainData.jsonl`)
- **Format**: Single unified JSON Lines file (`.jsonl`)
- **Local Path**: `rag_corpus/raw/vedas/continueousPreTrainData.jsonl`
- **Download / Access Instructions**: Included in local repo directory.
- **Preprocessing & Chunking Strategy**:
  - **Vedic Texts (Rig, Sama, Yajur, Atharva Veda)**: Chunked at Hymn (Sukta), Anuvaka, or Verse granularity.
  - **Ayurvedic Samhitas (Charaka, Sushruta, Rasa Jala Nidhi)**: Chunked at Chapter / Section level.
  - **IRJAY Research Papers**: Page-level chunking with 150-character overlap.
- **Train / Validation / Test Splits**: Single continuous pre-training / RAG retrieval split (~10K to ~100K records).
- **Columns / Features / Labels**:
  - `text`: Structured payload formatted with collection title, translator, chapter/sutra heading, and translation text payload.
- **Target Variable**: Unsupervised text / RAG context retrieval payload.
- **Where Used in Code**: Parsed by `backend/src/rag/parse.ts`, chunked in `backend/src/rag/chunk.ts`, embedded via Gemini API (`embed.ts`), and queried by `backend/src/rag/retriever.ts`.
- **Model / Task it Supports**: Classical Ayush RAG reasoning engine (`ayushRag.ts`), VedaGPT domain pre-training.
- **Citation**: Rig Veda (Ralph T.H. Griffith 1896), Black Yajur Veda (Arthur Berriedale Keith 1914), Sushruta Samhita (Kaviraj Kunja Lal Bhishagratna 1907), Rasa Jala Nidhi (Rasacharya Kaviraj Bhudeb Mookerji 1926).
- **Privacy / Ethical Notes**: Ancient historical and classical literature; no ethical or privacy constraints.